CSD 330 – Security Analytics

Homework 1

Part 2

1. Find a server on the University of Houston network. Explain how you can be sure that it is on UH network. Use the helpful program called nmap to find all the open ports and services running. Make a record of the services that are running.

ANSWER:

```
Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\kabir> start-transcript
Transcript started, output file is C:\Users\kabir\OneDrive\Documents\PowerShell_transcript.KABIRKANHASPECT.Nqc7jg7w.20200203171716.txt
PS C:\Users\kabir> resolve-dnsname uh.edu -type DNSKEY

Name                           Type TTL   Section    PrimaryServer              NameAdministrator          SerialNumber
----                           ---- ---   -------    -------------              -----------------          ------------
uh.edu                         SOA  899   Authority  uhgrid2.e.uh.edu           reza.uh.edu                2010015245


PS C:\Users\kabir> resolve-dnsname uh.edu -type DS

Name                           Type TTL   Section    PrimaryServer              NameAdministrator          SerialNumber
----                           ---- ---   -------    -------------              -----------------          ------------
edu                            SOA  900   Authority  a.edu-servers.net          nstld.verisign-grs.com     1580730517


PS C:\Users\kabir> resolve-dnsname uh.edu -type DNSKEY -DnssecOk

Name                           Type TTL   Section    PrimaryServer              NameAdministrator          SerialNumber
----                           ---- ---   -------    -------------              -----------------          ------------
uh.edu                         SOA  867   Authority  uhgrid2.e.uh.edu           reza.uh.edu                2010015245


PS C:\Users\kabir> nmap uh.edu
Starting Nmap 7.80 ( https://nmap.org ) at 2020-02-03 17:19 India Standard Time
Nmap scan report for uh.edu (129.7.97.54)
Host is up (0.27s latency).
rDNS record for 129.7.97.54: www.uh.edu
Not shown: 987 closed ports
PORT     STATE    SERVICE
53/tcp   filtered domain
80/tcp   open     http
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
443/tcp  open     https
445/tcp  filtered microsoft-ds
1433/tcp filtered ms-sql-s
1434/tcp filtered ms-sql-m
2000/tcp open     cisco-sccp
3389/tcp filtered ms-wbt-server
5060/tcp open     sip
8008/tcp open     http
8010/tcp open     xmpp
```

Verisign is a trusted certificate authority and since it has verified the server, a chain of trust is created and hence we have safely believe that this is indeed the server at University of Houston. Ports 80, 443, 2000, 5060, 8008 and 8010 are open.

2. Use NVD and CVE to find vulnerabilities in the services that are running. For this you will need the version numbers of each service on an open port. What options do you use for nmap?

```
Nmap done: 1 IP address (1 host up) scanned in 17.29 seconds
PS C:\Users\kabir> nmap -sV uh.edu
Starting Nmap 7.80 ( https://nmap.org ) at 2020-02-03 17:20 India Standard Time
Stats: 0:01:17 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 66.67% done; ETC: 17:22 (0:00:28 remaining)
Stats: 0:02:19 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 83.33% done; ETC: 17:23 (0:00:24 remaining)
Stats: 0:02:39 elapsed; 0 hosts completed (1 up), 1 undergoing Service Scan
Service scan Timing: About 100.00% done; ETC: 17:23 (0:00:00 remaining)
Stats: 0:03:16 elapsed; 0 hosts completed (1 up), 1 undergoing Script Scan
NSE Timing: About 91.67% done; ETC: 17:24 (0:00:01 remaining)
Nmap scan report for uh.edu (129.7.97.54)
Host is up (0.31s latency).
Not shown: 987 closed ports
PORT     STATE    SERVICE        VERSION
53/tcp   filtered domain
80/tcp   open     http
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
443/tcp  open     ssl/http       Apache httpd
445/tcp  filtered microsoft-ds
1433/tcp filtered ms-sql-s
1434/tcp filtered ms-sql-m
2000/tcp open     tcpwrapped
3389/tcp filtered ms-wbt-server
5060/tcp open     tcpwrapped
8008/tcp open     http
8010/tcp open     ssl/http-proxy FortiGate Web Filtering Service
```

Here, we see that we are unable to obtain any version numbers for the running services, hence we cannot use NVD or CVE to find vulnerabilities.

```
2 services unrecognized despite returning data. If you know the service/version, please submit the following fingerprints at https://nmap.org/cgi-bin/submit.cgi?new-service :
==============NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)==============
SF-Port80-TCP:V=7.80%I=7%D=2/3%Time=5E380941%P=i686-pc-windows-windows%r(H
SF:TTPOptions,F7,"HTTP/1\.0\x20307\x20Temporary\x20Redirect\r\nLocation:\x
SF:20https://uh\.edu/\r\nContent-Type:\x20text/html\r\nCache-Control:\x20p
SF:rivate\r\nConnection:\x20close\r\nDate:\x20Mon,\x2003\x20Feb\x202020\x2
SF:011:51:29\x20GMT\r\n\r\n<head><body>\x20This\x20object\x20may\x20be\x20
SF:found\x20<a\x20HREF=\"https://uh\.edu/\">here</a>\x20</body>")%r(RTSPRe
SF:quest,70,"HTTP/1\.1\x20200\x20OK\r\nContent-Type:\x20text/html\r\nPragm
SF:a:\x20no-cache\r\nRefresh:\x201;\x20URL=\r\nConnection:\x20Close\r\n\r\
SF:n<HTML></HTML>")%r(FourOhFourRequest,137,"HTTP/1\.0\x20302\x20Object\x2
SF:0Moved\r\nDate:\x20Mon,\x2003\x20Feb\x202020\x2011:51:34\x20GMT\r\nLoca
SF:tion:\x20https://uh\.edu/nice%20ports%2C/Tri%6Eity\.txt%2ebak\r\nConten
SF:t-Type:\x20text/html\r\nCache-Control:\x20private\r\nConnection:\x20clo
SF:se\r\n\r\n<head><body>\x20This\x20object\x20may\x20be\x20found\x20<a\x2
SF:0HREF=\"https://uh\.edu/nice%20ports%2C/Tri%6Eity\.txt%2ebak\">here</a>
SF:\x20</body>")%r(Help,D0,"HTTP/1\.1\x20307\x20Temporary\x20Redirect\r\nL
SF:ocation:\x20https://uh\.edu\r\nContent-Type:\x20text/html\r\nCache-Cont
SF:rol:\x20private\r\nConnection:\x20close\r\n\r\n<head><body>\x20This\x20
SF:object\x20may\x20be\x20found\x20<a\x20HREF=\"https://uh\.edu\">here</a>
SF:\x20</body>")%r(TerminalServerCookie,D0,"HTTP/1\.1\x20307\x20Temporary\
SF:x20Redirect\r\nLocation:\x20https://uh\.edu\r\nContent-Type:\x20text/ht
SF:ml\r\nCache-Control:\x20private\r\nConnection:\x20close\r\n\r\n<head><b
SF:ody>\x20This\x20object\x20may\x20be\x20found\x20<a\x20HREF=\"https://uh
SF:\.edu\">here</a>\x20</body>")%r(Kerberos,D0,"HTTP/1\.1\x20307\x20Tempor
SF:ary\x20Redirect\r\nLocation:\x20https://uh\.edu\r\nContent-Type:\x20tex
SF:t/html\r\nCache-Control:\x20private\r\nConnection:\x20close\r\n\r\n<hea
SF:d><body>\x20This\x20object\x20may\x20be\x20found\x20<a\x20HREF=\"https:
SF://uh\.edu\">here</a>\x20</body>")%r(SMBProgNeg,D0,"HTTP/1\.1\x20307\x20
SF:Temporary\x20Redirect\r\nLocation:\x20https://uh\.edu\r\nContent-Type:\
SF:x20text/html\r\nCache-Control:\x20private\r\nConnection:\x20close\r\n\r
SF:\n<head><body>\x20This\x20object\x20may\x20be\x20found\x20<a\x20HREF=\"
SF:https://uh\.edu\">here</a>\x20</body>");
==============NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)==============
SF-Port8008-TCP:V=7.80%I=7%D=2/3%Time=5E38093B%P=i686-pc-windows-windows%r
SF:(GetRequest,CC,"HTTP/1\.1\x20302\x20Found\r\nLocation:\x20https://:8010
SF:/\r\nConnection:\x20close\r\nX-Frame-Options:\x20SAMEORIGIN\r\nX-XSS-Pr
SF:otection:\x201;\x20mode=block\r\nX-Content-Type-Options:\x20nosniff\r\n
SF:Content-Security-Policy:\x20frame-ancestors\r\n\r\n")%r(FourOhFourReque
SF:st,EF,"HTTP/1\.1\x20302\x20Found\r\nLocation:\x20https://:8010/nice%20p
SF:orts%2C/Tri%6Eity\.txt%2ebak\r\nConnection:\x20close\r\nX-Frame-Options
SF::\x20SAMEORIGIN\r\nX-XSS-Protection:\x201;\x20mode=block\r\nX-Content-T
SF:ype-Options:\x20nosniff\r\nContent-Security-Policy:\x20frame-ancestors\
SF:r\n\r\n")%r(GenericLines,CB,"HTTP/1\.1\x20302\x20Found\r\nLocation:\x20
SF:https://:8010\r\nConnection:\x20close\r\nX-Frame-Options:\x20SAMEORIGIN
SF:\r\nX-XSS-Protection:\x201;\x20mode=block\r\nX-Content-Type-Options:\x2
SF:0nosniff\r\nContent-Security-Policy:\x20frame-ancestors\r\n\r\n")%r(HTT
SF:POptions,CB,"HTTP/1\.1\x20302\x20Found\r\nLocation:\x20https://:8010\r\
SF:nConnection:\x20close\r\nX-Frame-Options:\x20SAMEORIGIN\r\nX-XSS-Protec
SF:tion:\x201;\x20mode=block\r\nX-Content-Type-Options:\x20nosniff\r\nCont
SF:ent-Security-Policy:\x20frame-ancestors\r\n\r\n")%r(RTSPRequest,CB,"HTT
SF:P/1\.1\x20302\x20Found\r\nLocation:\x20https://:8010\r\nConnection:\x20
SF:close\r\nX-Frame-Options:\x20SAMEORIGIN\r\nX-XSS-Protection:\x201;\x20m
SF:ode=block\r\nX-Content-Type-Options:\x20nosniff\r\nContent-Security-Pol
SF:icy:\x20frame-ancestors\r\n\r\n")%r(SIPOptions,CB,"HTTP/1\.1\x20302\x20
SF:Found\r\nLocation:\x20https://:8010\r\nConnection:\x20close\r\nX-Frame-
SF:Options:\x20SAMEORIGIN\r\nX-XSS-Protection:\x201;\x20mode=block\r\nX-Co
SF:ntent-Type-Options:\x20nosniff\r\nContent-Security-Policy:\x20frame-anc
SF:estors\r\n\r\n");

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 198.06 seconds
```

3. Check Metasploit for exploit for the vulnerability and report what Metasploit returns.

This question is redundant, because, as mentioned earlier, we were unable to find any vulnerabilities.

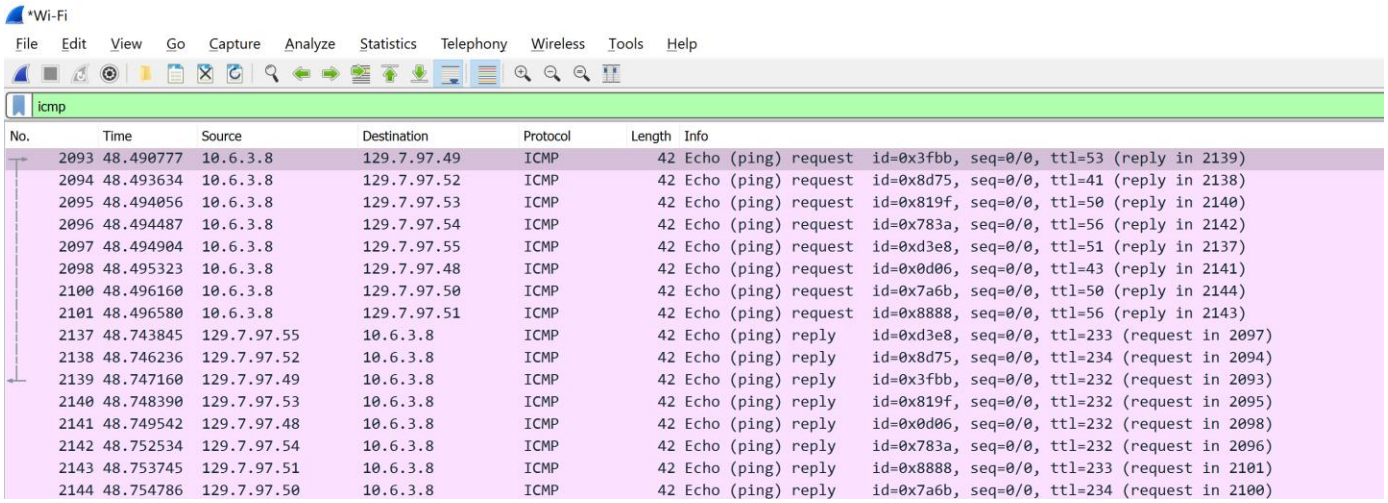
4. Next try Nmap on a small range of addresses that includes the server you found.

```
PS C:\Users\kabir> Resolve-DnsName uh.edu

Name                                           Type   TTL   Section    IPAddress
----                                           ----   ---   -------    ---------
uh.edu                                         A      10759 Answer     129.7.97.54


PS C:\Users\kabir> nmap 129.7.97.48/29
Starting Nmap 7.80 ( https://nmap.org ) at 2020-02-03 17:32 India Standard Time
Stats: 0:01:17 elapsed; 0 hosts completed (8 up), 8 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 56.88% done; ETC: 17:34 (0:00:58 remaining)
Stats: 0:02:00 elapsed; 0 hosts completed (8 up), 8 undergoing SYN Stealth Scan
SYN Stealth Scan Timing: About 85.92% done; ETC: 17:34 (0:00:20 remaining)
Nmap scan report for ramp.research.uh.edu (129.7.97.48)
Host is up (0.25s latency).
Not shown: 987 closed ports
PORT     STATE    SERVICE
53/tcp   filtered domain
80/tcp   open     http
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
443/tcp  open     https
445/tcp  filtered microsoft-ds
1433/tcp filtered ms-sql-s
1434/tcp filtered ms-sql-m
2000/tcp open     cisco-sccp
3389/tcp filtered ms-wbt-server
5060/tcp open     sip
8008/tcp open     http
8010/tcp open     xmpp

Nmap scan report for mysitedev.uh.edu (129.7.97.49)
Host is up (0.26s latency).
Not shown: 987 closed ports
PORT     STATE    SERVICE
53/tcp   filtered domain
80/tcp   open     http
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
443/tcp  open     https
445/tcp  filtered microsoft-ds
1433/tcp filtered ms-sql-s
1434/tcp filtered ms-sql-m
2000/tcp open     cisco-sccp
3389/tcp filtered ms-wbt-server
5060/tcp open     sip
8008/tcp open     http
8010/tcp open     xmpp
```

```
Nmap scan report for sharedev.uh.edu (129.7.97.50)
Host is up (0.26s latency).
Not shown: 987 closed ports
PORT     STATE    SERVICE
53/tcp   filtered domain
80/tcp   open     http
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
443/tcp  open     https
445/tcp  filtered microsoft-ds
1433/tcp filtered ms-sql-s
1434/tcp filtered ms-sql-m
2000/tcp open     cisco-sccp
3389/tcp filtered ms-wbt-server
5060/tcp open     sip
8008/tcp open     http
8010/tcp open     xmpp

Nmap scan report for mysite.uh.edu (129.7.97.51)
Host is up (0.25s latency).
Not shown: 987 closed ports
PORT     STATE    SERVICE
53/tcp   filtered domain
80/tcp   open     http
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
443/tcp  open     https
445/tcp  filtered microsoft-ds
1433/tcp filtered ms-sql-s
1434/tcp filtered ms-sql-m
2000/tcp open     cisco-sccp
3389/tcp filtered ms-wbt-server
5060/tcp open     sip
8008/tcp open     http
8010/tcp open     xmpp

Nmap scan report for share.uh.edu (129.7.97.52)
Host is up (0.25s latency).
Not shown: 987 closed ports
PORT     STATE    SERVICE
53/tcp   filtered domain
80/tcp   open     http
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
443/tcp  open     https
445/tcp  filtered microsoft-ds
1433/tcp filtered ms-sql-s
1434/tcp filtered ms-sql-m
2000/tcp open     cisco-sccp
3389/tcp filtered ms-wbt-server
5060/tcp open     sip
8008/tcp open     http
8010/tcp open     xmpp
```

```
Nmap scan report for auth.my.uh.edu (129.7.97.53)
Host is up (0.26s latency).
Not shown: 986 closed ports
PORT     STATE    SERVICE
53/tcp   filtered domain
80/tcp   open     http
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
443/tcp  open     https
445/tcp  filtered microsoft-ds
1433/tcp filtered ms-sql-s
1434/tcp filtered ms-sql-m
2000/tcp open     cisco-sccp
3389/tcp filtered ms-wbt-server
5060/tcp open     sip
8008/tcp open     http
8010/tcp open     xmpp
8443/tcp filtered https-alt

Nmap scan report for campusnet.uh.edu (129.7.97.54)
Host is up (0.25s latency).
Not shown: 987 closed ports
PORT     STATE    SERVICE
53/tcp   filtered domain
80/tcp   open     http
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
443/tcp  open     https
445/tcp  filtered microsoft-ds
1433/tcp filtered ms-sql-s
1434/tcp filtered ms-sql-m
2000/tcp open     cisco-sccp
3389/tcp filtered ms-wbt-server
5060/tcp open     sip
8008/tcp open     http
8010/tcp open     xmpp

Nmap scan report for ssl.uh.edu (129.7.97.55)
Host is up (0.25s latency).
Not shown: 987 closed ports
PORT     STATE    SERVICE
53/tcp   filtered domain
80/tcp   open     http
135/tcp  filtered msrpc
139/tcp  filtered netbios-ssn
443/tcp  open     https
445/tcp  filtered microsoft-ds
1433/tcp filtered ms-sql-s
1434/tcp filtered ms-sql-m
2000/tcp open     cisco-sccp
3389/tcp filtered ms-wbt-server
5060/tcp open     sip
8008/tcp open     http
8010/tcp open     xmpp
```

5. Using Wireshark and nmap together. The goal is to capture the nmap traffic using Wireshark (run Nmap in ping mode) and capture all the traffic for 3 seconds. Explain how you know that you are capturing the right traffic.



No.	Time	Source	Destination	Protocol	Length	Info
2093	48.490777	10.6.3.8	129.7.97.49	ICMP	42	Echo (ping) request id=0x3fbb, seq=0/0, ttl=53 (reply in 2139)
2094	48.493634	10.6.3.8	129.7.97.52	ICMP	42	Echo (ping) request id=0x8d75, seq=0/0, ttl=41 (reply in 2138)
2095	48.494056	10.6.3.8	129.7.97.53	ICMP	42	Echo (ping) request id=0x819f, seq=0/0, ttl=50 (reply in 2140)
2096	48.494487	10.6.3.8	129.7.97.54	ICMP	42	Echo (ping) request id=0x783a, seq=0/0, ttl=56 (reply in 2142)
2097	48.494904	10.6.3.8	129.7.97.55	ICMP	42	Echo (ping) request id=0xd3e8, seq=0/0, ttl=51 (reply in 2137)
2098	48.495323	10.6.3.8	129.7.97.48	ICMP	42	Echo (ping) request id=0x0d06, seq=0/0, ttl=43 (reply in 2141)
2100	48.496160	10.6.3.8	129.7.97.50	ICMP	42	Echo (ping) request id=0x7a6b, seq=0/0, ttl=50 (reply in 2144)
2101	48.496580	10.6.3.8	129.7.97.51	ICMP	42	Echo (ping) request id=0x8888, seq=0/0, ttl=56 (reply in 2143)
2137	48.743845	129.7.97.55	10.6.3.8	ICMP	42	Echo (ping) reply id=0xd3e8, seq=0/0, ttl=233 (request in 2097)
2138	48.746236	129.7.97.52	10.6.3.8	ICMP	42	Echo (ping) reply id=0x8d75, seq=0/0, ttl=234 (request in 2094)
2139	48.747160	129.7.97.49	10.6.3.8	ICMP	42	Echo (ping) reply id=0x3fbb, seq=0/0, ttl=232 (request in 2093)
2140	48.748390	129.7.97.53	10.6.3.8	ICMP	42	Echo (ping) reply id=0x819f, seq=0/0, ttl=232 (request in 2095)
2141	48.749542	129.7.97.48	10.6.3.8	ICMP	42	Echo (ping) reply id=0x0d06, seq=0/0, ttl=232 (request in 2098)
2142	48.752534	129.7.97.54	10.6.3.8	ICMP	42	Echo (ping) reply id=0x783a, seq=0/0, ttl=232 (request in 2096)
2143	48.753745	129.7.97.51	10.6.3.8	ICMP	42	Echo (ping) reply id=0x8888, seq=0/0, ttl=233 (request in 2101)
2144	48.754786	129.7.97.50	10.6.3.8	ICMP	42	Echo (ping) reply id=0x7a6b, seq=0/0, ttl=234 (request in 2100)

```
PS C:\Users\kabir> nmap 129.7.97.48/29 -sn
Starting Nmap 7.80 ( https://nmap.org ) at 2020-02-03 17:36 India Standard Time
Nmap scan report for ramp.research.uh.edu (129.7.97.48)
Host is up (0.0040s latency).
Nmap scan report for mysitedev.uh.edu (129.7.97.49)
Host is up (0.0020s latency).
Nmap scan report for sharedev.uh.edu (129.7.97.50)
Host is up (0.0030s latency).
Nmap scan report for mysite.uh.edu (129.7.97.51)
Host is up (0.0030s latency).
Nmap scan report for share.uh.edu (129.7.97.52)
Host is up (0.0020s latency).
Nmap scan report for auth.my.uh.edu (129.7.97.53)
Host is up (0.0030s latency).
Nmap scan report for m.uh.edu (129.7.97.54)
Host is up (0.0020s latency).
Nmap scan report for ssl.uh.edu (129.7.97.55)
Host is up (0.0040s latency).
Nmap done: 8 IP addresses (8 hosts up) scanned in 0.78 seconds
```

```
PS C:\Users\kabir> Get-NetIPAddress -InterfaceAlias Wi-Fi


IPAddress         : fe80::2424:8298:8387:3789%5
InterfaceIndex    : 5
InterfaceAlias    : Wi-Fi
AddressFamily     : IPv6
Type              : Unicast
PrefixLength      : 64
PrefixOrigin      : WellKnown
SuffixOrigin      : Link
AddressState      : Preferred
ValidLifetime     : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource      : False
PolicyStore       : ActiveStore

IPAddress         : 10.6.3.8
InterfaceIndex    : 5
InterfaceAlias    : Wi-Fi
AddressFamily     : IPv4
Type              : Unicast
PrefixLength      : 20
PrefixOrigin      : Dhcp
SuffixOrigin      : Dhcp
AddressState      : Preferred
ValidLifetime     : 1.11:37:42
PreferredLifetime : 1.11:37:42
SkipAsSource      : False
PolicyStore       : ActiveStore
```

We see that our IP address is the same as the one from whom the ping requests are originating and replies are coming to on wireshark. Hence, verified.

6. Use a password cracking program such as John the Ripper to find all vulnerable accounts on your own devices. Report the login names and passwords it found. Run it for at least 30 minutes on each device.

```
PS C:\Users\kabir> & "$x/john.exe" $(ls ./etc-shadow.txt)
Warning: detected hash type "HMAC-SHA256", but the string is also recognized as "HMAC-SHA512"
Use the "--format=HMAC-SHA512" option to force loading these as that type instead
Warning: only loading hashes of type "HMAC-SHA256", but also saw type "md5crypt"
Use the "--format=md5crypt" option to force loading hashes of that type instead
Warning: only loading hashes of type "HMAC-SHA256", but also saw type "md5crypt-long"
Use the "--format=md5crypt-long" option to force loading hashes of that type instead
Warning: only loading hashes of type "HMAC-SHA256", but also saw type "md5crypt-opencl"
Use the "--format=md5crypt-opencl" option to force loading hashes of that type instead
Using default input encoding: UTF-8
Loaded 1 password hash (HMAC-SHA256 [password is key, SHA256 256/256 AVX2 8x])
Will run 8 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Warning: Only 29 candidates buffered for the current salt, minimum 64 needed for performance.
Proceeding with wordlist:/run/password.lst, rules:Wordlist
Proceeding with incremental:ASCII
0g 0:00:00:57  3/3 0g/s 5956Kp/s 5956Kc/s 5956KC/s crienta144..curanathai
0g 0:00:01:12  3/3 0g/s 6554Kp/s 6554Kc/s 6554KC/s almotsk2..alos136a
0g 0:00:03:44  3/3 0g/s 8764Kp/s 8764Kc/s 8764KC/s esroiia..esdhkdh
0g 0:00:05:59  3/3 0g/s 8950Kp/s 8950Kc/s 8950KC/s culu20jm..culsmeio
0g 0:00:21:54  3/3 0g/s 9799Kp/s 9799Kc/s 9799KC/s fj6hai4..fp22825
0g 0:00:21:56  3/3 0g/s 9797Kp/s 9797Kc/s 9797KC/s hn1nm1c..hn1swg9
0g 0:00:21:58  3/3 0g/s 9795Kp/s 9795Kc/s 9795KC/s el ovy9..el 13g,
0g 0:00:22:01  3/3 0g/s 9798Kp/s 9798Kc/s 9798KC/s 41GSO41..41G0mru
0g 0:00:22:02  3/3 0g/s 9803Kp/s 9803Kc/s 9803KC/s jcz5o2..jczxxs
0g 0:00:22:04  3/3 0g/s 9801Kp/s 9801Kc/s 9801KC/s gu80z0..gu8ke.
0g 0:00:22:07  3/3 0g/s 9804Kp/s 9804Kc/s 9804KC/s B4gifk..B4gcor
0g 0:00:22:10  3/3 0g/s 9807Kp/s 9807Kc/s 9807KC/s chymycrun..chyeacl17
0g 0:00:24:43  3/3 0g/s 9697Kp/s 9697Kc/s 9697KC/s wf52tro..wf55sfr
0g 0:03:00:53  3/3 0g/s 3802Kp/s 3802Kc/s 3802KC/s cjpus454..cjput017
0g 0:03:21:34  3/3 0g/s 3627Kp/s 3627Kc/s 3627KC/s je&bpk..je&geh
0g 0:03:21:36  3/3 0g/s 3628Kp/s 3628Kc/s 3628KC/s tf-26A..tf-kji
0g 0:03:21:40  3/3 0g/s 3629Kp/s 3629Kc/s 3629KC/s 6iGJMZ..6yPhly
0g 0:03:21:42  3/3 0g/s 3630Kp/s 3630Kc/s 3630KC/s zs iq2..zs 50*
0g 0:03:21:43  3/3 0g/s 3631Kp/s 3631Kc/s 3631KC/s B2zlha..B2zqdt
0g 0:03:21:47  3/3 0g/s 3633Kp/s 3633Kc/s 3633KC/s sokawu21..sokyus82
```

I ran John the Ripper on a Linux password file for 3 hours! Still no concrete result was obtained.

7. Use openssl to generate a key pair and run encryption and decryption algorithms on a short text file.

Please see transcript file.